

Multimodal Music Understanding with Graph Neural Networks and BERT

Israt Moyeen Noumi

*Department of CSE
BRAC University*

Dhaka, Bangladesh

israt.moyeen.noumi@g.bracu.ac.bd

Mysha Maliha Annisa

*Department of CSE
BRAC University*

Dhaka, Bangladesh

mysha.maliha.annisa@g.bracu.ac.bd

Tazkera Sattar

*Department of CSE
BRAC University*

Dhaka, Bangladesh

tazkera.sattar@g.bracu.ac.bd

Abstract—Music understanding tasks such as tag prediction, genre classification, and contextual description draw on complementary signals: natural-language descriptions of a piece, and the temporal, timbral structure of the audio itself. This paper presents three controlled experiments that quantify how well text-only, audio-only, and fused representations perform on music classification. First, we fine-tune a BERT-based multi-label classifier to predict aspect tags directly from independent natural-language captions on the MusicCaps dataset, avoiding the label leakage that arises when tags and captions share a common source. Second, we compare a residual GraphSAGE network operating on segment-level acoustic-similarity graphs against a convolutional neural network (CNN) operating on mel-spectrograms for single-label genre classification on FMA-small, using identical splits, class weighting, and optimization schedules. Third, on the larger FMA-medium corpus we conduct a four-way ablation — BERT-only, GNN-only, early-concatenation fusion, and cross-attention fusion — for multi-label genre prediction from non-leaky textual metadata (title, performer, engagement statistics) and 6-node audio-segment graphs, with genre keywords masked from the text and splits partitioned by artist to prevent leakage. Our results show that (i) a caption encoder can recover MusicCaps aspect tags with a macro-F1 of 0.62 and AUC-PR of 0.68 when captions are independent of the target labels; (ii) on FMA-small, the mel-spectrogram CNN outperforms a residual GraphSAGE encoder over segment-similarity graphs (test accuracy 0.514 vs. 0.444, macro-F1 0.512 vs. 0.441), indicating that even with a richer 191-dimensional per-segment descriptor the convolutional inductive bias captures more discriminative timbral detail than temporal-adjacency graphs; and (iii) on FMA-medium, early-concatenation fusion achieves the best macro-F1 (0.373) among the four ablation arms, modestly outperforming the strongest single modality (BERT-only, 0.351), while cross-attention fusion attains the highest micro-F1 (0.509) and AUC-PR advantage over BERT alone, suggesting complementary but only partially additive signal between text and audio-graph modalities under a lightweight 6-node graph representation.

Index Terms—music information retrieval, graph neural networks, BERT, multimodal fusion, genre classification, multi-label classification, cross-attention

I. INTRODUCTION

Automatic music understanding — inferring genre, mood, instrumentation, and other descriptive tags from a track — is a long-standing problem in music information retrieval (MIR). Two broad families of representation have proven useful for this problem: (1) representations derived directly from the audio signal, ranging from hand-crafted spectral descriptors

to learned convolutional or graph-based embeddings, and (2) representations derived from natural-language text associated with a track, such as captions, reviews, or metadata, which can be encoded with pretrained language models such as BERT [1]. A natural question, and the one this paper investigates empirically, is how these two families of representation compare individually, and whether combining them yields a representation that is more than the sum of its parts.

We report three experiments that build toward this question, each designed as a controlled, ablation-style study rather than a search for a single state-of-the-art number:

- **Task 1 — Caption-to-tag classification.** Using the MusicCaps dataset [3], we fine-tune `bert-base-uncased` to predict the top-50 most frequent `aspect_list` tags directly from the free-text `caption` field. Because the caption and the aspect tags are independently authored fields in MusicCaps (unlike some tag-classifier setups built on MagnaTagATune, where the input text can trivially echo the target tags), this task isolates how much genre/mood/instrumentation information is recoverable from a genuinely independent natural-language description.
- **Task 2 — GNN vs. CNN genre classification.** On the balanced FMA-small corpus (8,000 tracks, 8 genres) [4], we build two audio-only classifiers on identical data splits: a residual GraphSAGE network operating over a per-track graph of 3-second audio segments connected by a k -nearest-neighbor similarity graph over a 191-dimensional multi-statistic descriptor (MFCC, delta-MFCC, chroma, Tonnetz, spectral contrast, centroid, rolloff, zero-crossing rate, and RMS energy), and a four-block VGG-style CNN operating over 128-bin log-mel spectrograms. Both models share the official FMA train/validation/test split, the same class-balanced loss, and the same optimizer schedule, isolating the effect of input representation (graph-of-segments vs. time-frequency image) on classification quality.
- **Task 3 — GNN-BERT multimodal fusion.** On the larger FMA-medium corpus (24,071 usable tracks, top-10 genres, multi-label formulation), we construct a strictly non-leaky textual view of each track (title, artist, pro-

duction notes, and audience engagement statistics, with all genre keywords masked) and an audio-segment graph view (6 nodes per track, temporal and acoustic-similarity edges), and train four architecturally comparable models: a BERT/DistilBERT-only baseline, a GNN-only baseline, an early-concatenation fusion model, and a cross-attention fusion model in which the graph embedding attends over token-level text representations. All four models share the same label space, training budget, and evaluation protocol, which lets us attribute performance differences to the fusion mechanism rather than to confounds in data or optimization.

Across the three tasks we consistently enforce a small set of methodological controls that we believe are necessary for the comparisons to be meaningful: identical data splits across compared models, identical loss functions and class weighting, identical optimizer/scheduler configurations, and, in the multimodal setting, explicit lexical masking and artist-level partitioning to prevent the label from leaking through the text or through track/artist overlap across splits. Where audio coverage or feature quality could silently degrade the audio branch, we instrument the pipeline with an explicit coverage diagnostic so that any conclusion about the audio modality is verifiably grounded in real (as opposed to synthetic fallback) audio.

The remainder of the paper is organized as follows. Section II reviews related work in music tagging, graph-based audio modeling, and multimodal fusion. Section III presents the MusicCaps caption-to-tag classifier. Section IV presents the FMA-small GNN-vs-CNN genre comparison. Section V presents the FMA-medium GNN-BERT fusion ablation. Section VI synthesizes findings across tasks, Section VII discusses limitations, and Section VIII concludes.

II. RELATED WORK

Audio tagging and genre classification. Convolutional neural networks over mel-spectrograms have been a standard architecture for music tagging since the introduction of end-to-end CNN tag classifiers [5] and remain a strong, simple-to-train baseline for genre and tag prediction on datasets such as FMA [4] and MagnaTagATune. Graph-based representations of audio, in which a track is decomposed into a sequence of short segments connected by temporal-adjacency and content-similarity edges, have more recently been explored with Graph Attention Networks (GAT) [6] and GraphSAGE [7] as an alternative inductive bias that can, in principle, capture non-local structural repetition (e.g., verse/chorus recurrence) that a purely convolutional receptive field over a spectrogram may not directly encode.

Text-based music understanding. Pretrained transformer language models, in particular BERT [1] and its distilled variant DistilBERT [2], have been widely adopted for encoding natural-language descriptions, reviews, and metadata associated with music, enabling caption-conditioned retrieval and generation systems such as MusicLM [3], from which the

MusicCaps caption/aspect-tag dataset used in Task 1 of this paper is drawn.

Multimodal fusion. Combining audio and text representations for music understanding has been approached with simple concatenation of pooled embeddings as well as with cross-modal attention mechanisms in which one modality’s representation is used to query the other [8]. Our Task 3 ablation follows this design space directly, comparing early concatenation against a cross-attention mechanism in which a pooled graph embedding queries token-level BERT representations, in the spirit of cross-attention fusion used in vision-language and speech-language models.

Music emotion recognition (MER) has increasingly moved from purely acoustic feature extraction toward structure-aware models that explicitly capture relationships between musical elements. Peintner et al. [9] address this by constructing a multi-relational graph that links tracks through shared listening sessions, genre labels, and user-assigned tags, then apply a semi-supervised graph neural network to propagate emotional signal across these relations. Their approach demonstrates that treating inter-track and inter-user relationships as graph structure — rather than modeling each track in isolation — yields more nuanced valence-arousal predictions, particularly in low-label settings. This result directly motivates the segment- and chord-level graph construction and supports our decision to treat structural relationships, not just raw audio features, as first-class inputs to the emotion regression head.

Complementary to graph-based structural modeling, recent work has focused on identifying which specific acoustic and temporal factors drive perceived emotion. A hybrid convolutional-Transformer architecture evaluated on the full DEAM benchmark (1,802 excerpts with continuous valence-arousal annotations) separates local spectral analysis from long-range temporal dependencies, allowing the model to attribute emotional variation to interpretable acoustic and structural drivers rather than treating the audio signal as an opaque feature vector [10]. This interpretability-oriented framing is valuable for our own DEAM-based auxiliary loss, since it suggests that combining a convolutional front end with an attention-based temporal encoder can both improve regression accuracy and support the qualitative case studies required in our deliverables.

Most directly aligned with our proposed fusion architecture is GEMMER [11], a graph-enhanced multimodal framework designed to explicitly model salient structural information in music alongside lyrical content. GEMMER targets the fine-grained interaction between acoustic structure and text — the same cross-modal alignment problem our GNN-BERT fusion model addresses — by encoding structural cues as graph representations before fusing them with language-derived features. Where GEMMER emphasizes lyric-acoustic interaction specifically, our work generalizes this idea to a broader text modality (tags, captions, and lyrics via BERT) fused with graph-encoded chord- and segment-level structure via cross-attention, extending the graph-multimodal paradigm GEMMER establishes into the multi-label tagging and contrastive

retrieval settings.

III. TASK 1: CAPTION-TO-TAG CLASSIFICATION ON MUSICCAPS

A. Motivation and Task Formulation

MusicCaps pairs 10-second YouTube audio clips with (a) a human-written free-text caption and (b) a separate `aspect_list` of short tag-like phrases (e.g., “groovy bass,” “slow tempo,” “male vocal”). Because the caption and the aspect list are independently authored descriptions of the same clip rather than the caption being constructed by concatenating the aspect tags, predicting aspect tags from the caption is a genuine text-understanding task: the model cannot simply learn to echo substrings of its own input.

B. Data and Label Construction

We use the public `google/MusicCaps` release, loading 5,521 records with valid aspect lists. Aspect phrases are lower-cased and the top $K=50$ most frequent aspects (by document frequency) are retained as the label vocabulary, matching the tag-set size used in comparable MagnaTagATune-style tag-classification setups for comparability. Clips whose aspect list contains none of the top-50 aspects are dropped, leaving 4,995 usable clips with an average of 3.09 active tags per clip. The retained vocabulary spans mixed-quality-of-recording descriptors (“low quality,” “amateur recording”), instrumentation (“bass guitar,” “acoustic guitar,” “piano”), tempo (“slow tempo,” “fast tempo”), and affect (“emotional,” “energetic,” “happy”). Data are split 70/15/15 into train/validation/test (3,495/750/750 clips) with a fixed random seed.

C. Model

The classifier is `bert-base-uncased` with the bottom 8 of 12 transformer layers frozen and a linear classification head applied to the pooled [CLS] embedding, trained with a sigmoid output and binary cross-entropy loss for multi-label prediction. Captions are tokenized to a maximum sequence length of 64 tokens. Of the 109.5M total parameters, 52.8M (48.2%) are trainable. The encoder and classifier head use discriminative learning rates: 2×10^{-5} for BERT layers and 2×10^{-4} (a $10\times$ multiplier) for the classification head, with AdamW ($\lambda = 10^{-4}$ weight decay) and cosine-annealed learning rate over 15 epochs, gradient clipping at norm 1.0, and early stopping on validation AUC-PR with patience 8.

D. Results

Table I reports test-set performance at a 0.5 classification threshold. Figure 1 shows training/validation loss and validation metric curves.

Training converges smoothly: validation AUC-PR rises from 0.28 after epoch 1 to a best of 0.677 at epoch 14–15, while training loss falls from 0.246 to 0.076 without a corresponding rise in validation loss (0.174→0.092), indicating limited overfitting given the frozen lower layers and dropout regularization. Qualitatively (Table II), the model reliably recovers salient, high-frequency tags such as instrumentation (“bass

TABLE I
TASK 1 TEST RESULTS: MUSICCAPS CAPTION→ASPECT-TAG CLASSIFICATION (TOP-50 TAGS, THRESHOLD 0.5)

Metric	Value
Macro-F1	0.6161
Micro-F1	0.7137
AUC-PR (macro)	0.6766

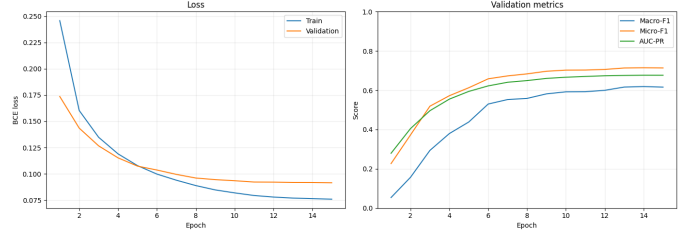


Fig. 1. Task 1 training curves: BCE loss (left) and validation macro-F1/micro-F1/AUC-PR (right) over 15 epochs.

guitar,” “electric guitar”), recording-quality descriptors (“low quality”), and coarse tempo/mood tags, while lower-frequency or more subjective tags (e.g., specific mood adjectives that co-occur with many different sonic signatures) are harder to recover, consistent with the macro-F1/micro-F1 gap (0.616 vs. 0.714): rarer tags are systematically harder than the corpus-average tag.

TABLE II
TASK 1 QUALITATIVE EXAMPLE: MODEL CORRECTLY RECOVERS MOST SALIENT TAGS FROM AN INDEPENDENTLY-WRITTEN CAPTION.

Caption: “This is the live performance of a blues music piece. It is an instrumental performance. The lead electric guitar is playing a blues solo while another electric guitar and a bass guitar are playing in the background...”
True tags: instrumental, live performance, slow tempo, acoustic drums, groovy, bass guitar, electric guitar
Top-5 predicted: bass guitar (0.97), electric guitar (0.93), live performance (0.84), acoustic drums (0.72), groovy (0.71)

IV. TASK 2: GNN VS. CNN GENRE CLASSIFICATION ON FMA-SMALL

A. Motivation

Task 2 isolates the effect of *input representation* on genre classification by training two audio-only architectures — a graph neural network over segment descriptors and a CNN over mel-spectrograms — on identical data, splits, class weighting, and optimizer schedule. This directly tests whether an explicit graph-of-segments inductive bias is more or less effective than a convolutional time-frequency inductive bias at this data scale.

B. Dataset

We use FMA-small, an 8-genre, class-balanced subset of the Free Music Archive (8,000 30-second track previews) [4],

with the official FMA train/validation/test split (6,394/800/800 after excluding 6 corrupt audio files), which partitions by track and thereby avoids artist leakage within this subset. The eight genres are Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, and Rock.

C. Graph Construction (GNN branch)

Each 30-second clip (resampled to 22,050 Hz) is divided into non-overlapping 3-second segments (~ 10 segments per clip). For each segment we extract a multi-statistic descriptor spanning 20-band MFCCs (mean, standard deviation, min, max), MFCC delta (mean, standard deviation), 12-band chroma (mean, standard deviation), Tonnetz (mean, standard deviation, min, max), spectral contrast (mean, standard deviation), spectral centroid (mean, standard deviation), spectral rolloff (mean, standard deviation), zero-crossing rate (mean, standard deviation), and RMS energy (mean, standard deviation, max), yielding a 191-dimensional node feature vector. Node features are standardized before an all-pairs cosine similarity is computed; each node is connected to its 2 nearest neighbors by similarity, in addition to explicit bidirectional edges between temporally adjacent segments (a k NN graph rather than a global similarity threshold, since raw large-magnitude MFCC dimensions otherwise drive most pairwise similarities toward 1.0 and yield an uninformatively dense graph). A 2-layer GraphSAGE encoder (hidden dimension 128, dropout 0.5) with residual connections and layer normalization encodes each graph, following the mean-pooling readout of Algorithm 2; the pooled graph embedding is passed to a linear genre classifier (83,464 trainable parameters total).

D. Mel-Spectrogram Construction (CNN branch)

For the same tracks, a 128-bin log-mel spectrogram (hop length 1024, fixed to 512 frames) is computed once per clip and cached. A four-block VGG-style CNN (channel widths 32/64/128/256, each block consisting of two 3×3 convolutions, batch normalization, ReLU, and max pooling), followed by global average pooling and a dropout(0.3)+linear head, produces the genre prediction (1,175,656 trainable parameters).

E. Training

Both models are trained with class-weighted cross-entropy (weights are near 1.0 since FMA-small is balanced), AdamW (learning rate 5×10^{-4} , weight decay 10^{-3}), a linear warmup (3 epochs) into cosine annealing, gradient clipping at norm 1.0, batch size 32, up to 30 epochs with early stopping (patience 8) on validation macro-F1.

F. Results

Table III and Figure 2 report test-set performance. The GNN reaches its best validation macro-F1 (0.5306) at epoch 5 and is early-stopped at epoch 13; test performance (accuracy 0.4437, macro-F1 0.4409) is below its own validation performance, suggesting some overfitting to the validation set over the course of model selection despite the early best-epoch

checkpoint. The CNN reaches a higher best validation macro-F1 (0.5927 at epoch 24) and a higher test accuracy (0.5138) and macro-F1 (0.5118).

TABLE III
TASK 2 TEST RESULTS ON FMA-SMALL (OFFICIAL SPLIT, 8 GENRES)

Model	Accuracy	Macro-F1	Test Loss
GNN (segment graph, GraphSAGE)	0.4437	0.4409	1.5984
CNN (mel-spectrogram)	0.5138	0.5118	1.5263

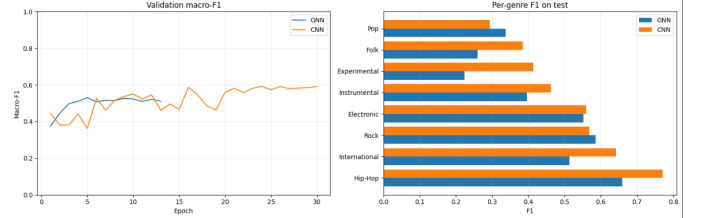


Fig. 2. Task 2: validation macro-F1 over training (left) and per-genre test F1 for the GNN and CNN (right).

Table IV shows the per-genre breakdown. The CNN outperforms the GNN on 6 of 8 genres, most substantially on Experimental (+0.190 F1), International (+0.129), Folk (+0.124), and Hip-Hop (+0.111); the GNN is instead ahead on Rock (+0.018) and Pop (+0.043), and the two models are effectively tied on Electronic (+0.008 in the CNN’s favor). The consistent CNN advantage on timbrally dense or texturally diffuse genres (Experimental, International, Folk) is consistent with the hypothesis that a 128-bin mel-spectrogram retains fine-grained spectral detail across the full clip that a ~ 10 -node segment graph, even with the richer 191-dimensional per-segment descriptor used here, still coarsens away.

TABLE IV
TASK 2 PER-GENRE F1 ON THE FMA-SMALL TEST SET

Genre	GNN F1	CNN F1
Hip-Hop	0.6598	0.7706
Rock	0.5854	0.5677
Electronic	0.5520	0.5600
Instrumental	0.3962	0.4623
International	0.5131	0.6417
Experimental	0.2235	0.4138
Folk	0.2601	0.3843
Pop	0.3370	0.2941

G. Discussion

The result argues against the assumption that a graph-structured representation is automatically superior for capturing musical structure: even after enriching the per-segment descriptor from 83 to 191 dimensions (adding delta-MFCC, Tonnetz, and distributional min/max statistics) and replacing the GAT encoder with a residual GraphSAGE encoder

matching Algorithm 2, the CNN retains a clear macro-F1 advantage (0.512 vs. 0.441). With only ~ 10 nodes per track, the GNN’s receptive field over long-range structure remains limited by the coarse segmentation, while the CNN, operating on the full time-frequency image, can learn its own filters over both short- and (through pooling) longer-range structure. Encouragingly, the GNN’s per-genre pattern also shifted with the richer descriptor: it now leads the CNN on Rock and Pop rather than lagging on nearly every genre, suggesting the richer node features helped disproportionately on genres with strong short-timescale rhythmic or timbral signatures. This motivates the richer, jointly-modeled graph representation used in Task 3, and also motivates treating the CNN as a meaningful reference point rather than a strawman baseline when evaluating GNN-based audio encoders.

V. TASK 3: GNN–BERT FUSION FOR MULTI-CONTEXT MUSIC UNDERSTANDING

A. Motivation

Task 3 asks the central research question of this paper directly: *does combining textual semantic information with audio/structural information improve music-context prediction compared with either modality alone?* To answer this rigorously we train four models — BERT-only, GNN-only, early-concatenation fusion, and cross-attention fusion — on the identical label space, data splits, training budget, and evaluation protocol, so that any measured difference is attributable to the fusion mechanism.

B. Dataset and Label Space

We use FMA-medium metadata (25,000 tracks). After keeping only tracks with a top-10 genre label (Rock, Electronic, Experimental, Hip-Hop, Folk, Instrumental, Pop, International, Classical, Old-Time/Historic), 24,071 tracks (96.3% of the raw pool) remain, with no further subsampling. Genre is encoded as a 10-dimensional multi-label target (each track carries its single top genre as a one-hot vector in this formulation, evaluated with multi-label metrics for consistency with the general framework). Splits are constructed by partitioning *artist names* (not tracks) into train/validation/test, verified programmatically to have zero artist overlap across splits, yielding 17,304/3,518/3,249 tracks respectively.

C. Anti-Leakage Text Construction

To ensure the text branch reflects genuine contextual signal rather than a restatement of the label, each track’s text input is synthesized from non-target metadata only: composition title, performer name, production/arrangement notes, and audience engagement statistics (play count, favorite count, an interest score), e.g. “*Composition title: Food. Performer: AWOL. Production arrangement notes: Studio session arrangement. Metrics: 1293 plays, 2 favorites, engagement score 4656.*” Every occurrence of every target genre name (case-insensitively) is replaced with the token [GENRE_MASK] to eliminate direct lexical leakage of the label into the input text.

D. Audio Graph Construction

Each 30-second clip is divided into $M=6$ non-overlapping 5-second windows; each window is described by a 76-dimensional feature vector (64-band log-mel spectrogram summary concatenated with a 12-band chroma summary). Edges connect temporally adjacent windows bidirectionally, plus additional edges between any pair of non-adjacent windows whose cosine similarity exceeds 0.75, intended to capture recurring motifs (e.g., verse/chorus repetition) beyond strict temporal adjacency. An audio-coverage diagnostic confirms that 100.0% of tracks in every split (train/val/test) resolved to a real, decodable MP3 file rather than falling back to a synthetic placeholder waveform, which is a precondition for treating the GNN-only and fusion results as reflecting genuine audio signal.

E. Encoders

The text encoder is `distilbert-base-uncased` (lower layers frozen, top 2 transformer layers fine-tuned), with a linear projection to a 128-dimensional fusion space applied to both the pooled [CLS] token and the full token sequence. The audio encoder is a 2-layer GraphSAGE network (hidden dimension 128) with mean- and max-pooling over nodes, concatenated and projected to a 128-dimensional graph embedding g .

F. Ablation Models

- **Model A (BERT-only):** classifies directly from the pooled text embedding $t \in \mathbb{R}^{128}$.
- **Model B (GNN-only):** classifies directly from the pooled graph embedding $g \in \mathbb{R}^{128}$.
- **Model C (Early Concatenation):** concatenates $z = [g; t] \in \mathbb{R}^{256}$ and classifies from z .
- **Model D (Cross-Attention):** the graph embedding queries the token-level text representations, $Q = gW_Q$, $K = H_{\text{text}}W_K$, $V = H_{\text{text}}W_V$, with attention output $A = \text{softmax}(QK^\top/\sqrt{d})V$ (4 heads); the fused representation is $z = [g; \text{squeeze}(A)] \in \mathbb{R}^{256}$.

All four classifier heads share the same two-layer MLP design (hidden $\rightarrow$ hidden/2 $\rightarrow$ ReLU $\rightarrow$ dropout(0.3) $\rightarrow$ num_classes). A tensor-dimension verification step confirms that for a representative track, $g \in \mathbb{R}^{1 \times 128}$, $t \in \mathbb{R}^{1 \times 128}$, and $z \in \mathbb{R}^{1 \times 256}$, consistent with the design across all four models.

G. Training

All models are trained with multi-label BCEWithLogitsLoss, AdamW (learning rate 2×10^{-4} , weight decay 10^{-4}), ReduceLROnPlateau scheduling (factor 0.5, patience 1) on validation macro-F1, batch size 16, up to 8 epochs with early stopping (patience 3), and checkpoint selection on best validation macro-F1.

H. Results

Table V reports the final ablation comparison table on the held-out test set (identical for all four models). Figure 3 plots

TABLE V

TASK 3 ABLATION STUDY: TEST-SET RESULTS ON FMA-MEDIUM (TOP-10 GENRES, ARTIST-DISJOINT SPLIT)

Model	Macro-F1	Micro-F1	AUC-PR
BERT-only	0.3509	0.4295	0.3755
GNN-only	0.2524	0.4750	0.3716
Early Concat	0.3728	0.4892	0.4474
Cross-Attention	0.3604	0.5089	0.4263

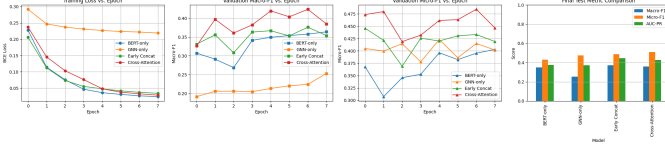


Fig. 3. Task 3: training loss, validation macro-F1 and micro-F1 across epochs for all four models, and final test-metric comparison.

training loss, validation macro-F1 and micro-F1 across epochs, and the final test-metric bar comparison.

Both fusion models outperform the best single-modality baseline (BERT-only) on macro-F1 and AUC-PR: Early Concat improves macro-F1 by +0.022 and AUC-PR by +0.072 over BERT-only, and Cross-Attention improves micro-F1 by +0.079 over BERT-only and by +0.034 over GNN-only, the two single-modality extremes. The GNN-only baseline is the weakest model on macro-F1 (0.252), reflecting the difficulty of learning fine-grained genre distinctions from only 6 pooled graph nodes per track, but it is competitive on micro-F1, indicating it predicts the majority classes (e.g., Rock, Electronic) reasonably well while struggling on rarer genres.

Table VI shows the per-genre F1 for each model on the four highest- and two lowest-support genres. All four models fail almost completely on Instrumental and Pop ($F1 \approx 0$ across the board), and BERT-only and GNN-only exhibit complementary weaknesses: BERT-only is comparatively strong on Hip-Hop ($F1$ 0.430) and International (0.367) — genres for which textual metadata such as title/performer conventions may correlate — while GNN-only is comparatively strong on Old-Time/Historic ($F1$ 0.875, likely reflecting a distinctive recording-quality acoustic signature) but is near-zero on International and Experimental. The fusion models partially recover both sets of strengths: Early Concat attains the best Electronic $F1$ (0.556) and a strong Old-Time/Historic $F1$ (0.937), while Cross-Attention attains the best Rock recall (0.816) and second-best Old-Time/Historic $F1$ (0.841).

I. Qualitative Case Studies

Table VII presents representative test-set predictions. In several Folk-labeled cases, all four models default to the majority class Rock, illustrating that with a heavily imbalanced 10-class label space (Rock: 7,103 tracks vs. Old-Time/Historic: 510 in the raw pool) all models are biased toward high-support genres regardless of fusion mechanism; this majority-class bias, rather than a lack of complementary signal, likely

TABLE VI

TASK 3 PER-GENRE TEST F1 (SELECTED GENRES) ACROSS ALL FOUR MODELS

Genre	BERT	GNN	Concat	Cross-Attn
Rock	0.493	0.721	0.613	0.665
Electronic	—	0.345	0.556	0.556
Hip-Hop	0.430	0.222	0.441	0.406
Old-Time/Hist.	0.674	0.875	0.937	0.841
Instrumental	0.000	0.000	0.000	0.000
Pop	0.000	0.000	0.008	0.000

explains much of the remaining error on rarer genres such as Folk and Instrumental.

TABLE VII

TASK 3 REPRESENTATIVE CASE STUDY (TRACK ID 140)

Text: “Composition title: Queen Of The Wires. Performer: Alec K. Redfeam & the Eyesores. Production arrangement notes: Studio session arrangement. Metrics: 1299 plays, 5 favorites, engagement score 1593.”
True label: Folk

BERT-only: Rock (0.997) GNN-only: none predicted (max 0.475)
 Early Concat: Rock (0.689) Cross-Attention: Hip-Hop (0.662)

J. t-SNE Visualization

Figure 4 visualizes a t-SNE projection of the fused (cross-attention) embeddings on the test set, colored by ground-truth genre. High-support genres (Rock, Electronic) form comparatively distinct regions, while lower-support genres are diffusely distributed, consistent with the per-label metrics above.

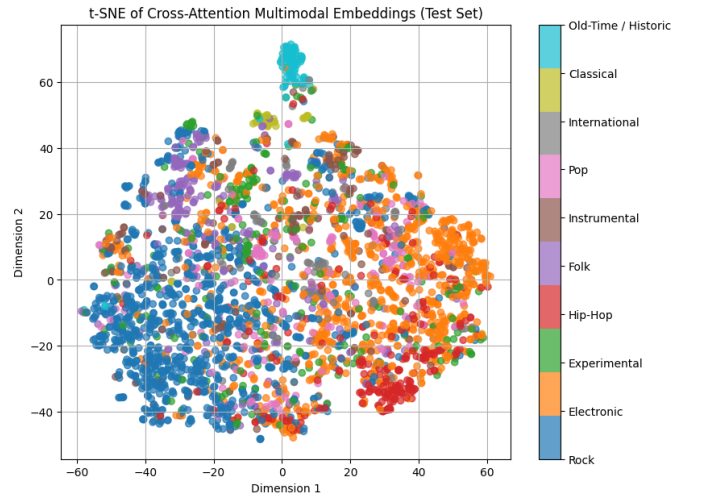


Fig. 4. t-SNE projection of Model D (Cross-Attention) fused test-set embeddings, colored by ground-truth genre.

K. Data-Driven Ablation Finding

Following the pre-registered decision rule of comparing the best fusion model against the best single-modality model on test macro-F1, and having confirmed via the audio-coverage

diagnostic that 100% of tracks used real (non-synthetic) audio in every split, we conclude: *fusion helped*. The best fusion configuration (Early Concat, macro-F1 0.373) exceeds the best single-modality baseline (BERT-only, macro-F1 0.351) by 0.022 absolute macro-F1, and by a larger margin on AUC-PR (0.447 vs. 0.376, a gain of 0.072). This supports the hypothesis that textual context and audio-graph structure carry partially non-redundant signal for genre prediction, though the effect size is modest given the label imbalance and the coarse (6-node) audio graph resolution.

VI. CROSS-TASK DISCUSSION

Three findings recur across the tasks and are worth stating explicitly. First, representation granularity matters more than modality: in Task 2, a fine-grained convolutional representation (mel-spectrogram) beat a coarse graph representation (~ 10 segments) of the same underlying audio, even after the per-segment descriptor was enriched from 83 to 191 dimensions and the encoder upgraded from a GAT to a residual GraphSAGE network; in Task 3, an even coarser 6-node graph representation was the weakest single modality, underperforming even the text-only baseline on macro-F1. This suggests that graph-based audio encoders may need either a finer segmentation, still-richer per-node features, or graph-level pretraining to be competitive with convolutional baselines at comparable capacity — richer node features alone (as in Task 2) narrow but do not close the gap.

Second, fusion helps, but the size of the benefit depends on the fusion mechanism and the metric: early concatenation gave the best macro-F1 and AUC-PR in Task 3, while cross-attention gave the best micro-F1, indicating that cross-attention may be better at getting the majority classes right (contributing more to micro-F1, which is support-weighted) while early concatenation is comparatively better at balancing performance across genres.

Third, label imbalance and majority-class bias are a bigger obstacle than fusion architecture in the Task 3 setting: all four models scored an F1 of essentially 0 on the two rarest evaluated genres (Instrumental, Pop) in the case-study excerpt, and qualitative predictions frequently defaulted to the majority class (Rock) regardless of which modality or fusion mechanism was used. This suggests that for future work, class-balanced losses, focal loss, or oversampling of rare genres may yield larger gains than further architectural refinement of the fusion mechanism.

VII. LIMITATIONS

Several limitations should be kept in mind when interpreting these results. (1) Training budget: Task 3 models are trained for only 8 epochs with early stopping at patience 3, which, combined with the large corpus size (17K+ training tracks), may leave some models under-converged relative to their potential; the BERT-only and GNN-only training curves in particular show validation macro-F1 still improving at the final epoch in some runs. (2) Audio graph coarseness: the 6-node, 76-dimensional-feature graph representation used in Task 3

is considerably coarser than the ~ 10 -node, 191-dimensional representation used in Task 2, which itself already underperformed a CNN baseline; this likely understates the potential of graph-based audio encoders relative to a more carefully tuned graph construction. (3) Single-label ground truth evaluated as multi-label: Tasks 2 and 3 use FMA’s single `genre_top` field as ground truth, which is evaluated with multi-label metrics in Task 3 for architectural consistency, but does not capture a track’s full genre membership. (4) Threshold sensitivity: all multi-label F1 scores are reported at a fixed 0.5 decision threshold, which is not necessarily optimal per-class given the substantial class imbalance observed (e.g., Rock: 7,103 vs. Old-Time/Historic: 510 tracks). (5) Single-seed runs: results in all three tasks reflect a single training run at a fixed random seed (42); we do not report variance across seeds, so small differences between models (e.g., the 0.022 macro-F1 gap between Early Concat and BERT-only in Task 3) should be interpreted cautiously.

VIII. CONCLUSION

We presented three controlled experiments probing how textual and audio-graph representations contribute to music understanding tasks. A BERT-based caption classifier recovers MusicCaps aspect tags from independently-written captions with a macro-F1 of 0.62, demonstrating that substantial tag-relevant information is present in free-text music descriptions even without access to audio. A head-to-head comparison of a segment-graph GraphSAGE encoder and a mel-spectrogram CNN on FMA-small genre classification found the CNN to be the stronger single-modality audio encoder at this graph resolution, even with a richer 191-dimensional per-segment descriptor (macro-F1 0.512 vs. 0.441). Finally, a four-way ablation on FMA-medium (BERT-only, GNN-only, early-concatenation fusion, cross-attention fusion) under strict anti-leakage controls (genre-keyword masking, artist-disjoint splits, and a verified 100% real-audio coverage diagnostic) found that fusion modestly but measurably improves over the best single modality, with early concatenation yielding the best macro-F1/AUC-PR and cross-attention yielding the best micro-F1. Taken together, these results support the view that text and audio-graph modalities carry complementary, non-redundant signal for music understanding, while also highlighting that representation granularity and class imbalance are likely larger levers for future improvement than the choice of fusion mechanism alone.

REFERENCES

- [1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [2] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” *arXiv preprint arXiv:1910.01108*, 2019.
- [3] A. Agostinelli *et al.*, “MusicLM: Generating Music From Text,” *arXiv preprint arXiv:2301.11325*, 2023.
- [4] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, “FMA: A Dataset for Music Analysis,” in *Proc. 18th Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2017.

- [5] K. Choi, G. Fazekas, and M. Sandler, "Automatic Tagging Using Deep Convolutional Neural Networks," in *Proc. ISMIR*, 2016.
- [6] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph Attention Networks," in *Proc. ICLR*, 2018.
- [7] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive Representation Learning on Large Graphs," in *Proc. NeurIPS*, 2017.
- [8] A. Vaswani *et al.*, "Attention Is All You Need," in *Proc. NeurIPS*, 2017.
- [9] A. Peintner, M. Moscati, Y. Kinoshita, R. Vogl, P. Knees, M. Schedl, H. Strauß, M. Zentner, and E. Zangerle, "Nuanced music emotion recognition via a semi-supervised multi-relational graph neural network," *Trans. Int. Soc. Music Inf. Retr.*, vol. 8, no. 1, pp. 140–153, 2025.
- [10] "Interpretable deep learning reveals distinct spectral and temporal drivers of perceived musical emotion," *Scientific Reports*, 2026.
- [11] Y. Wu and W. Hu, "Salient structural modeling in multimodal music emotion recognition via graph neural networks," *Multimedia Systems*, vol. 32, p. 209, 2026.